

Zeal College of Engineering and Research

Subject: LP-VI

Name: Jenil Girish Rathod

Class: BE

Div: B

Batch: B1

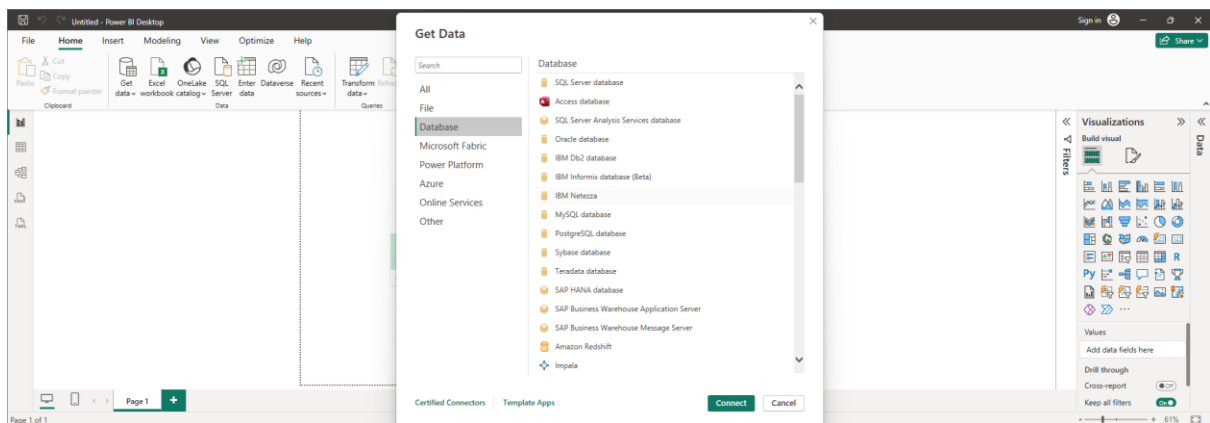
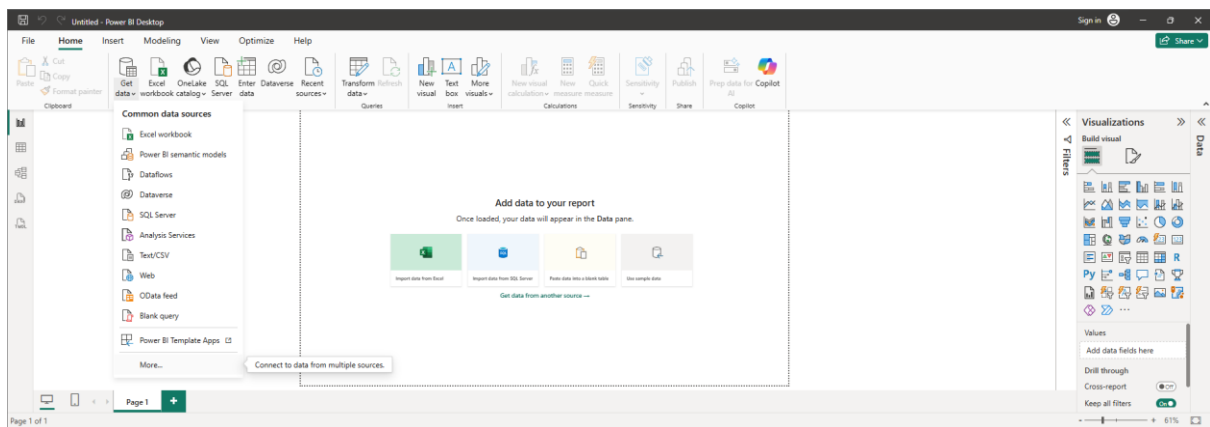
Roll No: B22009

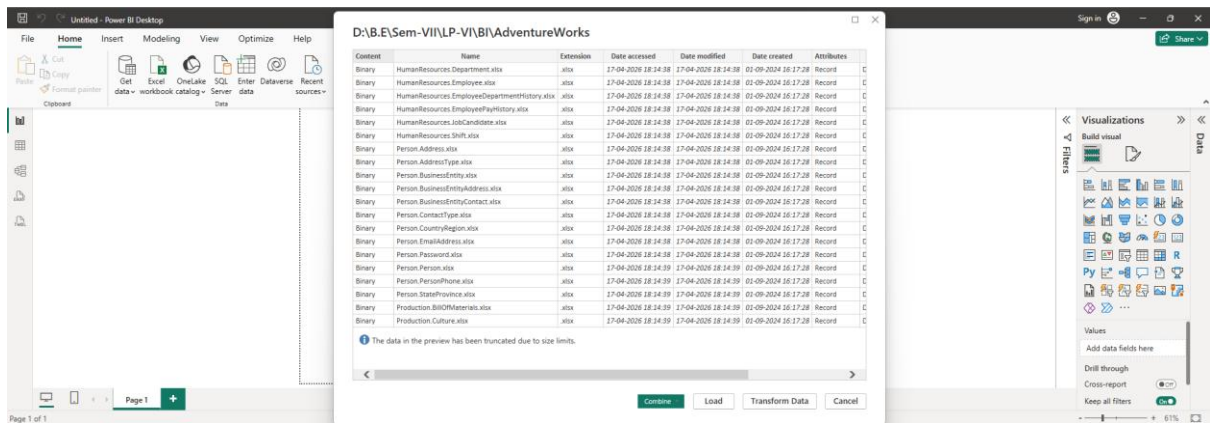
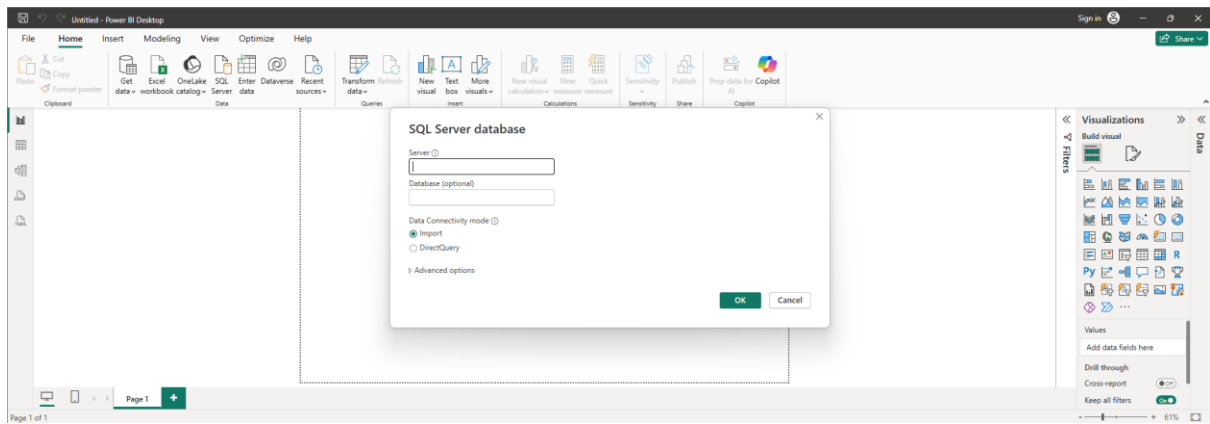
Group B: Practical No:2

Program Statement: Perform the Extraction Transformation and Loading (ETL) process to construct the database in the Sql server.

STEP 1: Open Power BI

1. Open Power BI Desktop
2. Click **Get Data** → **Excel**

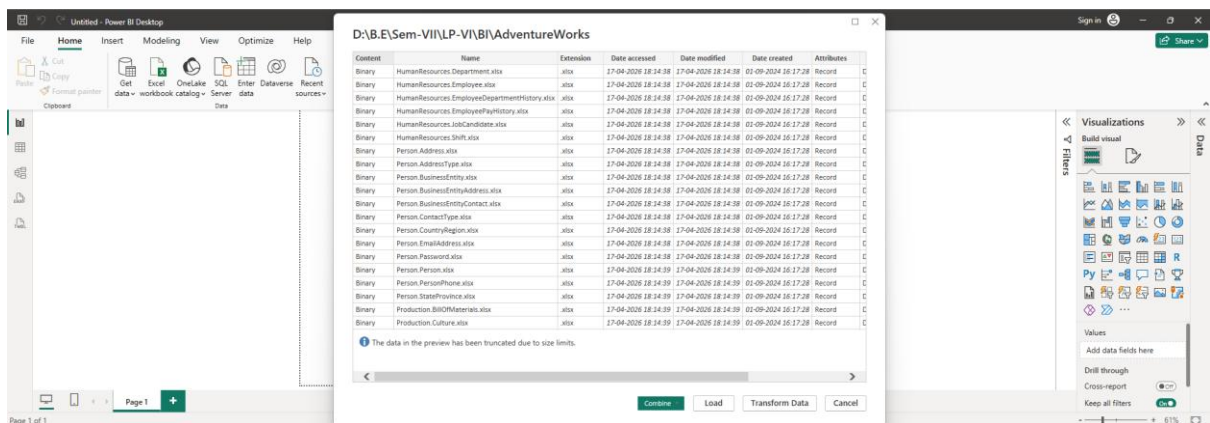




STEP 2: Extract Data (E in ETL)

1. Select your Excel file
2. Choose table/sheet
3. Click **Transform Data** (IMPORTANT)

This loads data into Power Query (staging layer)



Power Query Editor interface showing a table with columns: Content, Extension, Date acquired, Date modified, Date created, Attributes, and Folder Path. The table contains 33 rows of data, including entries for HumanResources, Production, and Purchasing.

Query Settings: Properties (Name: AdventureWorks), Applied Steps (Source).

8 COLUMNS, 33 ROWS Column profiling based on top 1000 rows

Power Query Editor interface showing a table with columns: DepartmentID, Name, GroupName, and ModifiedDate. The table contains 17 rows of data, including entries for Engineering, Sales, Marketing, Purchasing, Research and Development, Production, Human Resources, Finance, Information Services, Document Control, Quality Assurance, Facilities and Maintenance, Shipping and Receiving, and Executive.

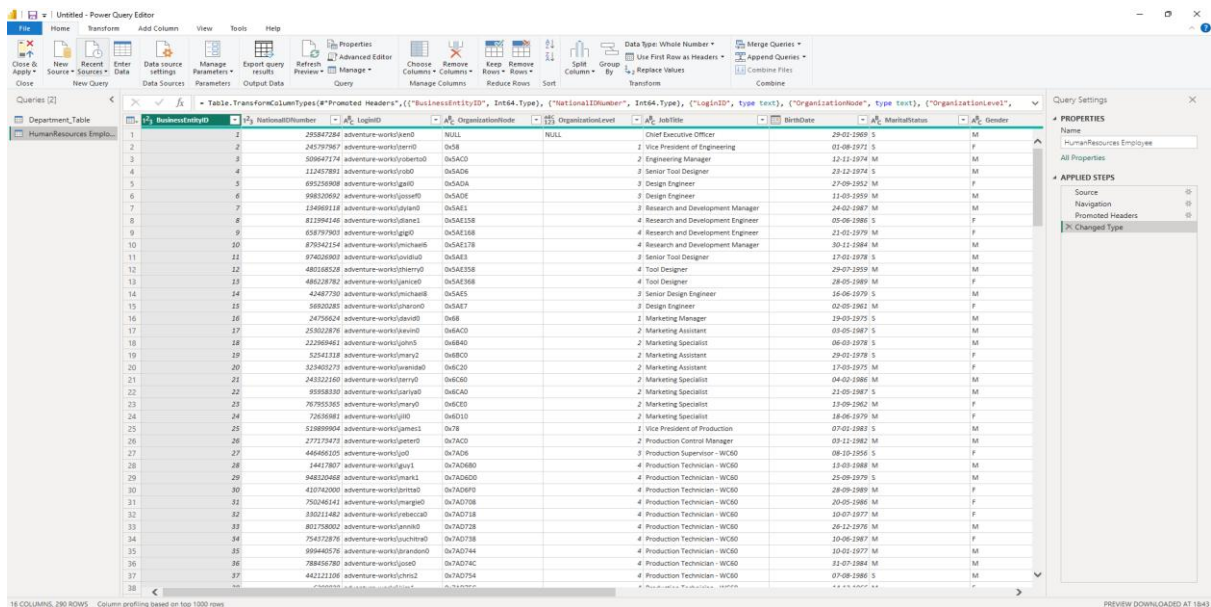
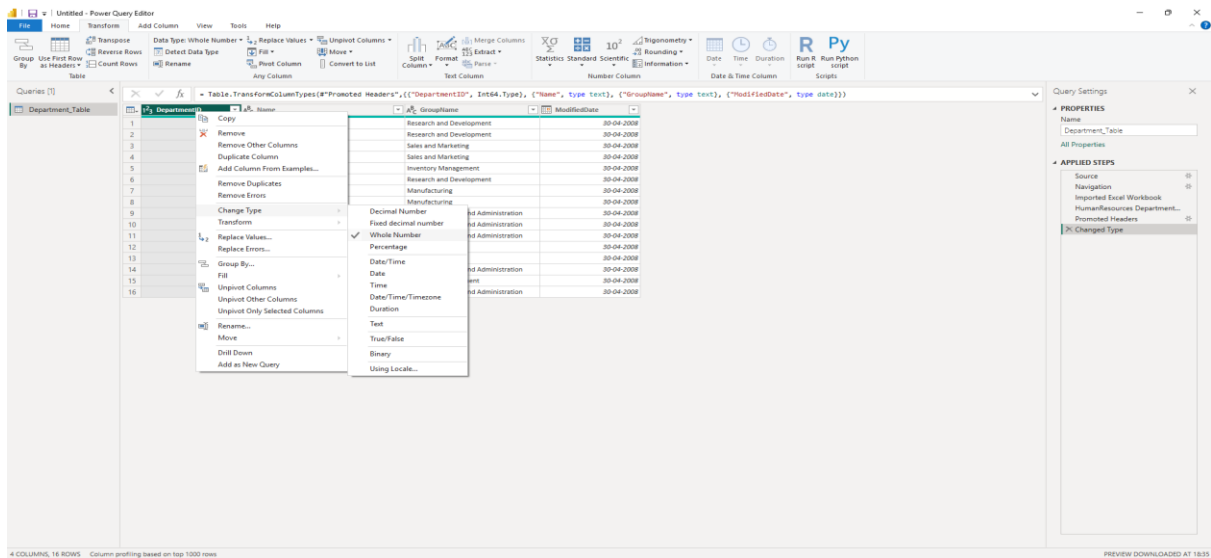
Query Settings: Properties (Name: AdventureWorks), Applied Steps (Navigation, Imported Excel Workbook, HumanResources Department...).

4 COLUMNS, 17 ROWS Column profiling based on top 1000 rows

Power Query Editor interface showing a table with columns: DepartmentID, Name, GroupName, and ModifiedDate. The table contains 16 rows of data, including entries for Engineering, Sales, Marketing, Purchasing, Research and Development, Production, Human Resources, Finance, Information Services, Document Control, Quality Assurance, Facilities and Maintenance, Shipping and Receiving, and Executive.

Query Settings: Properties (Name: AdventureWorks), Applied Steps (Navigation, Imported Excel Workbook, HumanResources Department..., Promoted Headers, Changed Type).

4 COLUMNS, 16 ROWS Column profiling based on top 1000 rows



STEP 3: Clean Data (Transform Step 1)

In Power Query:

Remove unwanted columns:

1. rowguid
2. OrganizationNode
3. LoginID
4. NationalIDNumber

Right-click → **Remove**

Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Report query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Reduce Rows Sort Split Columns Group By Data Type: Any * Merge Queries * Append Queries * Replace Values * Combine Files * Combine

Queries [2] Department_Table HumanResources.Employee

Table.TransformColumnTypes(*Promoted Headers*,{"BusinessEntityID", Int64.Type}, {"NationalIDNumber", Int64.Type}, {"LoginID", type text}, {"OrganizationNode", type text}, {"OrganizationLevel",

BusinessEntityID	NationalIDNumber	LoginID	JobTitle	BirthDate	MaritalStatus	Gender
1	295847284	adventure-works\jane0	Chief Executive Officer	29-01-1969	S	M
2	245797967	adventure-works\pam0	2 Vice President of Engineering	02-08-1972	S	F
3	509697176	adventure-works\patem0	2 Engineering Manager	12-12-1974	M	M
4	112457801	adventure-works\paul0	3 Senior Tool Designer	29-12-1974	S	M
5	695256908	adventure-works\gail0	3 Design Engineer	27-09-1952	M	F
6	988200652	adventure-works\jocse0	3 Design Engineer	11-03-1959	M	M
7	214969118	adventure-works\lylan0	3 Research and Development Manager	24-02-1987	M	M
8	813994145	adventure-works\clane1	4 Research and Development Engineer	05-06-1986	S	F
9	686797601	adventure-works\gail0	4 Research and Development Engineer	12-02-1979	M	F
10	876342154	adventure-works\micha5	4 Research and Development Manager	30-12-1984	M	M
11	574029901	adventure-works\jovita0	3 Senior Tool Designer	17-01-1978	S	M
12	480168528	adventure-works\pherry0	3 Tool Designer	29-07-1959	M	M
13	486228782	adventure-works\janice0	4 Tool Designer	28-05-1989	M	F
14	424877750	adventure-works\micha8	3 Senior Design Engineer	26-06-1979	S	M
15	569232361	adventure-works\paul0	3 Design Engineer	02-05-1981	M	F
16	24750624	adventure-works\dwain0	1 Marketing Manager	19-03-1975	S	M
17	253022876	adventure-works\kevin0	2 Marketing Assistant	09-05-1987	S	M
18	222694461	adventure-works\lored	2 Marketing Specialist	06-03-1978	S	M
19	325411218	adventure-works\mary2	2 Marketing Assistant	29-03-1978	S	F
20	523484375	adventure-works\paul0	2 Marketing Assistant	17-03-1975	M	F
21	243122260	adventure-works\jerry0	2 Marketing Specialist	04-02-1986	M	M
22	85958330	adventure-works\janis0	2 Marketing Specialist	21-05-1987	S	M
23	767953363	adventure-works\mary0	2 Marketing Specialist	13-09-1962	M	F
24	72635882	adventure-works\lilo	2 Marketing Specialist	18-06-1979	M	F
25	513895994	adventure-works\james1	1 Vice President of Production	07-01-1983	S	M
26	217215472	adventure-works\paul0	2 Production Control Manager	09-10-1962	M	F
27	444661010	adventure-works\jo0	3 Production Supervisor - WC60	08-10-1956	S	F
28	144173807	adventure-works\gus1	4 Production Technician - WC60	13-03-1968	M	M
29	948320468	adventure-works\mark1	4 Production Technician - WC60	25-09-1979	S	M
30	410742000	adventure-works\paul0	4 Production Technician - WC60	28-09-1989	M	F
31	792048142	adventure-works\margo0	4 Production Technician - WC60	20-05-1986	M	F
32	536211492	adventure-works\paul0	4 Production Technician - WC60	20-07-1977	M	F
33	801758002	adventure-works\jane0	4 Production Technician - WC60	26-12-1976	M	M
34	754372876	adventure-works\juchita0	4 Production Technician - WC60	10-06-1987	M	F
35	999440576	adventure-works\brandon0	4 Production Technician - WC60	10-01-1977	M	M
36	784567801	adventure-works\jane0	4 Production Technician - WC60	19-07-1984	M	M
37	442111106	adventure-works\jane1	4 Production Technician - WC60	07-08-1986	S	F
38						

16 COLUMNS, 280 ROWS Column profiling based on top 1000 rows

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Untitled - Power Query Editor

File Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Report query results Refresh Preview Advanced Editor Properties Choose Columns Remove Columns Keep Rows Reduce Rows Sort Split Columns Group By Data Type: Whole Number * Merge Queries * Append Queries * Replace Values * Combine Files * Combine

Queries [2] Department_Table HumanResources.Employee

Table.RemoveColumns(*Changed Type*,{"NationalIDNumber", "LoginID", "OrganizationNode", "rowguid", "ModifiedDate"})

Department	BusinessEntityID	OrganizationLevel	JobTitle	BirthDate	MaritalStatus	Gender	Salary	VacationHours
1	1	NULL	Chief Executive Officer	29-01-1969	S	M	14-01-2009	1
2	2	2	2 Vice President of Engineering	02-08-1972	S	F	23-02-2008	1
3	3	3	2 Engineering Manager	12-12-1974	M	M	11-12-2007	1
4	4	4	3 Senior Tool Designer	29-12-1974	S	M	05-12-2007	0
5	5	5	3 Design Engineer	27-09-1952	M	F	06-01-2008	1
6	6	6	3 Design Engineer	11-03-1959	M	M	24-02-2008	1
7	7	7	3 Research and Development Manager	24-02-1987	M	M	08-02-2009	1
8	8	8	4 Research and Development Engineer	05-06-1986	S	F	29-12-2008	1
9	9	9	4 Research and Development Engineer	12-02-1979	M	F	26-05-2009	1
10	10	10	4 Research and Development Manager	30-12-1984	M	M	05-05-2009	1
11	11	11	3 Senior Tool Designer	17-01-1978	S	M	05-12-2010	0
12	12	12	4 Tool Designer	29-07-1959	M	M	11-12-2007	0
13	13	13	4 Tool Designer	28-05-1989	M	F	23-12-2010	0
14	14	14	3 Senior Design Engineer	26-06-1979	S	M	30-12-2010	1
15	15	15	3 Design Engineer	02-05-1981	M	F	38-01-2011	1
16	16	16	1 Marketing Manager	19-03-1975	S	M	20-12-2007	1
17	17	17	2 Marketing Assistant	09-05-1987	S	M	26-01-2007	0
18	18	18	2 Marketing Specialist	06-03-1978	S	M	07-02-2011	0
19	19	19	2 Marketing Assistant	29-03-1978	S	F	14-02-2011	0
20	20	20	2 Marketing Assistant	17-03-1975	M	F	07-03-2011	0
21	21	21	2 Marketing Specialist	04-02-1986	M	M	03-03-2009	0
22	22	22	2 Marketing Specialist	21-05-1987	S	M	12-12-2008	0
23	23	23	2 Marketing Specialist	13-09-1962	M	F	12-03-2009	0
24	24	24	2 Marketing Specialist	18-06-1979	M	F	18-01-2009	0
25	25	25	1 Vice President of Production	07-01-1983	S	M	03-02-2009	1
26	26	26	2 Production Control Manager	09-10-1962	M	M	01-12-2008	1
27	27	27	3 Production Supervisor - WC60	08-10-1956	S	F	17-03-2008	0
28	28	28	4 Production Technician - WC60	13-03-1968	M	M	10-06-2006	0
29	29	29	4 Production Technician - WC60	25-09-1979	S	M	23-01-2009	0
30	30	30	4 Production Technician - WC60	28-09-1989	M	F	29-01-2009	0
31	31	31	4 Production Technician - WC60	20-05-1986	M	F	04-01-2009	0
32	32	32	4 Production Technician - WC60	10-07-1977	M	F	29-12-2008	0
33	33	33	4 Production Technician - WC60	26-12-1976	M	M	17-12-2008	0
34	34	34	4 Production Technician - WC60	10-06-1987	M	F	16-02-2009	0
35	35	35	4 Production Technician - WC60	10-01-1977	M	M	08-02-2009	0
36	36	36	4 Production Technician - WC60	19-07-1984	M	M	10-02-2009	0
37	37	37	4 Production Technician - WC60	07-08-1986	S	F	07-09-2009	0
38								

11 COLUMNS, 280 ROWS Column profiling based on top 1000 rows

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STEP 4: Handle Null Values

1. Filter out null rows
OR
2. Replace nulls with:
 - “Unknown” (text)
 - 0 (numbers)

Power Query Editor - Query: HumanResources.Employee

Table: HumanResources.Employee

Columns: BusinessEntityID, OrganizationLevel, JobTitle, BirthDate, MaritalStatus, Gender, HireDate, SalariedFlag, VacationHours

Query Settings: Name: HumanResources.Employee, Properties: All Properties, Applied Steps: Source, Navigation, Promoted Headers, Changed Type, Removed Columns

Replace Values dialog: Replace one value with another in the selected columns. Value To Find: NULL, Replace With: 0. OK, Cancel

11 COLUMNS, 290 ROWS. Column profiling based on top 1000 rows. PREVIEW DOWNLOADED AT 1843

Power Query Editor - Query: Department_Table

Table: Department_Table

Columns: DepartmentID, DepartmentName, JobTitle, BirthDate, MaritalStatus, Gender, HireDate, SalariedFlag, VacationHours

Query Settings: Name: Department_Table, Properties: All Properties, Applied Steps: Source, Navigation, Promoted Headers, Changed Type, Replaced Value

11 COLUMNS, 290 ROWS. Column profiling based on top 1000 rows. PREVIEW DOWNLOADED AT 1843

STEP 5: Split Address Column (Transform Step 2)

1. Select **JobTitle**
2. Click **Split Column → By Delimiter**
3. Choose:
 - Delimiter = Space
 - Split at = Left-most

You get:

- Role
- Department

11 Columns, 280 Rows. Column profiling based on top 1000 rows.

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Department_Table	BusinessEntityID	OrganizationLevel	JobTitle	StartDate	MaritalStatus	Gender	HireDate	SalaryFlag	VacationHours
1	1	1	Chief Executive Officer	29-01-2009	S	M	24-01-2009	1	0
2	2	2	Vice President of Engineering	01-08-1971	S	F	31-01-2008	1	0
3	3	3	Engineering Manager	12-12-1974	M	M	11-12-2007	1	0
4	4	4	Senior Tool Designer	27-09-1952	M	F	09-12-2007	0	0
5	5	5	Design Engineer	27-09-1952	M	F	06-01-2008	1	0
6	6	6	Design Engineer	11-03-1959	M	M	24-01-2009	1	0
7	7	7	Research and Development Manager	24-02-1987	M	M	08-02-2009	1	0
8	8	8	Research and Development Engineer	05-06-1986	S	F	29-12-2008	1	0
9	9	9	Research and Development Engineer	21-03-1979	M	F	18-01-2009	1	0
10	10	10	Research and Development Manager	30-11-1984	M	M	03-05-2009	1	0
11	11	11	Senior Tool Designer	17-01-1978	S	M	09-12-2010	0	0
12	12	12	Tool Designer	29-07-1959	M	M	11-12-2007	0	0
13	13	13	Tool Designer	28-05-1989	M	F	29-12-2010	0	0
14	14	14	Senior Design Engineer	16-06-1979	S	M	30-12-2010	1	0
15	15	15	Design Engineer	01-05-1982	M	F	18-01-2011	1	0
16	16	16	Marketing Manager	19-03-1975	S	M	10-12-2007	1	0
17	17	17	Marketing Assistant	03-05-1987	S	M	26-01-2007	0	0
18	18	18	Marketing Specialist	06-03-1978	S	M	07-02-2011	0	0
19	19	19	Marketing Assistant	29-01-1979	S	F	14-01-2011	0	0
20	20	20	Marketing Assistant	17-03-1975	M	F	07-02-2011	0	0
21	21	21	Marketing Specialist	04-02-1986	M	M	02-05-2009	0	0
22	22	22	Marketing Specialist	21-05-1987	S	M	12-12-2008	0	0
23	23	23	Marketing Specialist	13-09-1962	M	F	12-01-2009	0	0
24	24	24	Marketing Specialist	18-06-1979	M	F	18-01-2009	0	0
25	25	25	Vice President of Production	07-01-1982	S	M	03-02-2009	1	0
26	26	26	Production Control Manager	01-11-1982	M	M	01-12-2008	1	0
27	27	27	Production Supervisor - WCOG	08-10-1956	S	F	27-02-2008	0	0
28	28	28	Production Technician - WCOG	13-03-1988	M	M	30-06-2006	0	0
29	29	29	Production Technician - WCOG	25-09-1979	S	M	23-01-2009	0	0
30	30	30	Production Technician - WCOG	28-09-1989	M	F	29-01-2009	0	0
31	31	31	Production Technician - WCOG	20-05-1989	M	F	04-01-2009	0	0
32	32	32	Production Technician - WCOG	10-07-1977	M	F	29-12-2008	0	0
33	33	33	Production Technician - WCOG	26-12-1976	M	M	17-12-2008	0	0
34	34	34	Production Technician - WCOG	10-06-1987	M	F	16-02-2009	0	0
35	35	35	Production Technician - WCOG	10-01-1977	M	M	08-02-2009	0	0
36	36	36	Production Technician - WCOG	31-07-1984	M	M	10-02-2009	0	0
37	37	37	Production Technician - WCOG	07-08-1986	S	M	07-03-2009	0	0
38	38	38	Production Technician - WCOG	11-12-1997	S	F	11-12-2007	0	0

11 Columns, 280 Rows. Column profiling based on top 1000 rows.

REVIEW DOWNLOADED AT 1843

Department_Table	BusinessEntityID	OrganizationLevel	JobTitle	StartDate	MaritalStatus	Gender	HireDate	SalaryFlag	VacationHours
1	1	1	Chief Executive Officer	29-01-2009	S	M	24-01-2009	1	0
2	2	2	Vice President of Engineering	01-08-1971	S	F	31-01-2008	1	0
3	3	3	Engineering Manager	12-12-1974	M	M	11-12-2007	1	0
4	4	4	Senior Tool Designer	27-09-1952	M	F	09-12-2007	0	0
5	5	5	Design Engineer	27-09-1952	M	F	06-01-2008	1	0
6	6	6	Design Engineer	11-03-1959	M	M	24-01-2009	1	0
7	7	7	Research and Development Manager	24-02-1987	M	M	08-02-2009	1	0
8	8	8	Research and Development Engineer	05-06-1986	S	F	29-12-2008	1	0
9	9	9	Research and Development Engineer	21-03-1979	M	F	18-01-2009	1	0
10	10	10	Research and Development Manager	30-11-1984	M	M	03-05-2009	1	0
11	11	11	Senior Tool Designer	17-01-1978	S	M	09-12-2010	0	0
12	12	12	Tool Designer	29-07-1959	M	M	11-12-2007	0	0
13	13	13	Tool Designer	28-05-1989	M	F	29-12-2010	0	0
14	14	14	Senior Design Engineer	16-06-1979	S	M	30-12-2010	1	0
15	15	15	Design Engineer	01-05-1982	M	F	18-01-2011	1	0
16	16	16	Marketing Manager	19-03-1975	S	M	10-12-2007	1	0
17	17	17	Marketing Assistant	03-05-1987	S	M	26-01-2007	0	0
18	18	18	Marketing Specialist	06-03-1978	S	M	07-02-2011	0	0
19	19	19	Marketing Assistant	29-01-1979	S	F	14-01-2011	0	0
20	20	20	Marketing Assistant	17-03-1975	M	F	07-02-2011	0	0
21	21	21	Marketing Specialist	04-02-1986	M	M	02-05-2009	0	0
22	22	22	Marketing Specialist	21-05-1987	S	M	12-12-2008	0	0
23	23	23	Marketing Specialist	13-09-1962	M	F	12-01-2009	0	0
24	24	24	Marketing Specialist	18-06-1979	M	F	18-01-2009	0	0
25	25	25	Vice President	07-01-1982	S	M	03-02-2009	1	0
26	26	26	Production Control Manager	01-11-1982	M	M	01-12-2008	1	0
27	27	27	Production Supervisor - WCOG	08-10-1956	S	F	27-02-2008	0	0
28	28	28	Production Technician - WCOG	13-03-1988	M	M	30-06-2006	0	0
29	29	29	Production Technician - WCOG	25-09-1979	S	M	23-01-2009	0	0
30	30	30	Production Technician - WCOG	28-09-1989	M	F	29-01-2009	0	0
31	31	31	Production Technician - WCOG	20-05-1989	M	F	04-01-2009	0	0
32	32	32	Production Technician - WCOG	10-07-1977	M	F	29-12-2008	0	0
33	33	33	Production Technician - WCOG	26-12-1976	M	M	17-12-2008	0	0
34	34	34	Production Technician - WCOG	10-06-1987	M	F	16-02-2009	0	0
35	35	35	Production Technician - WCOG	10-01-1977	M	M	08-02-2009	0	0
36	36	36	Production Technician - WCOG	31-07-1984	M	M	10-02-2009	0	0
37	37	37	Production Technician - WCOG	07-08-1986	S	M	07-03-2009	0	0
38	38	38	Production Technician - WCOG	11-12-1997	S	F	11-12-2007	0	0

12 Columns, 280 Rows. Column profiling based on top 1000 rows.

REVIEW DOWNLOADED AT 1843

Department_Table	BusinessEntityID	OrganizationLevel	JobTitle_1	JobTitle_2	StartDate	MaritalStatus	Gender	HireDate	SalaryFlag	VacationHours
1	1	1	Chief Executive Officer		29-01-2009	S	M	24-01-2009	1	0
2	2	2	Vice President of Engineering		01-08-1971	S	F	31-01-2008	1	0
3	3	3	Engineering Manager		12-12-1974	M	M	11-12-2007	1	0
4	4	4	Senior Tool Designer		27-09-1952	M	F	09-12-2007	0	0
5	5	5	Design Engineer		27-09-1952	M	F	06-01-2008	1	0
6	6	6	Design Engineer		11-03-1959	M	M	24-01-2009	1	0
7	7	7	Research and Development Manager		24-02-1987	M	M	08-02-2009	1	0
8	8	8	Research and Development Engineer		05-06-1986	S	F	29-12-2008	1	0
9	9	9	Research and Development Engineer		21-03-1979	M	F	18-01-2009	1	0
10	10	10	Research and Development Manager		30-11-1984	M	M	03-05-2009	1	0
11	11	11	Senior Tool Designer		17-01-1978	S	M	09-12-2010	0	0
12	12	12	Tool Designer		29-07-1959	M	M	11-12-2007	0	0
13	13	13	Tool Designer		28-05-1989	M	F	29-12-2010	0	0
14	14	14	Senior Design Engineer		16-06-1979	S	M	30-12-2010	1	0
15	15	15	Design Engineer		01-05-1982	M	F	18-01-2011	1	0
16	16	16	Marketing Manager		19-03-1975	S	M	10-12-2007	1	0
17	17	17	Marketing Assistant		03-05-1987	S	M	26-01-2007	0	0
18	18	18	Marketing Specialist		06-03-1978	S	M	07-02-2011	0	0
19	19	19	Marketing Assistant		29-01-1979	S	F	14-01-2011	0	0
20	20	20	Marketing Assistant		17-03-1975	M	F	07-02-2011	0	0
21	21	21	Marketing Specialist		04-02-1986	M	M	02-05-2009	0	0
22	22	22	Marketing Specialist		21-05-1987	S	M	12-12-2008	0	0
23	23	23	Marketing Specialist		13-09-1962	M	F	12-01-2009	0	0
24	24	24	Marketing Specialist		18-06-1979	M	F	18-01-2009	0	0
25	25	25	Vice President of Production		07-01-1982	S	M	03-02-2009	1	0
26	26	26	Production Control Manager		01-11-1982	M	M	01-12-2008	1	0
27	27	27	Production Supervisor - WCOG		08-10-1956	S	F	27-02-2008	0	0
28	28	28	Production Technician - WCOG		13-03-1988	M	M	30-06-2006	0	0
29	29	29	Production Technician - WCOG		25-09-1979	S	M	23-01-2009	0	0
30	30	30	Production Technician - WCOG		28-09-1989	M	F	29-01-2009	0	0
31	31	31	Production Technician - WCOG		20-05-1989	M	F	04-01-2009	0	0
32	32	32	Production Technician - WCOG		10-07-1977	M	F	29-12-2008	0	0
33	33	33	Production Technician - WCOG		26-12-1976	M	M	17-12-2008	0	0
34	34	34	Production Technician - WCOG		10-06-1987	M	F	16-02-2009	0	0
35	35	35	Production Technician - WCOG		10-01-1977	M	M	08-02-2009	0	0
36	36	36	Production Technician - WCOG		31-07-1984	M	M	10-02-2009	0	0
37	37	37	Production Technician - WCOG		07-08-1986	S	M	07-03-2009	0	0
38	38	38	Production Technician - WCOG		11-12-1997	S	F	11-12-2007	0	0

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Queries [2]

Department_Table

HumanResources.Employee

Table: RenameColumns(ChangedType1,({"JobTitle.2", "Role"}, {"JobTitle.1", "Department"}))

BusinessEntityID	OrganizationLevel	Department	Role	BirthDate	MaritalStatus	Gender	HireDate	SalaryFlag
1	0	Chief	Executive Officer	29-01-1969	S	M	14-01-2009	
2	1	Vice	President of Engineering	05-09-1972	S	F	31-01-2008	
3	2	Engineering	Manager	12-12-1974	M	M	12-12-2007	
4	3	Senior	Tool Designer	29-12-1974	S	M	05-12-2007	
5	3	Design	Engineer	27-09-1992	M	F	06-01-2008	
6	3	Design	Engineer	12-03-1959	M	M	24-01-2008	
7	3	Research and Development	Manager	24-02-1987	M	M	08-02-2009	
8	4	Research and Development	Engineer	05-06-1986	S	F	29-12-2008	
9	4	Research and Development	Manager	21-03-1979	M	F	18-01-2009	
10	10	Research and Development	Manager	30-12-1984	M	M	03-05-2009	
11	11	Senior	Tool Designer	17-01-1978	S	M	05-12-2010	
12	12	Tool	Designer	29-07-1959	M	M	11-12-2007	
13	13	Tool	Designer	28-09-1989	M	F	29-12-2010	
14	14	Senior	Design Engineer	18-06-1979	S	M	30-12-2010	
15	15	Design	Engineer	03-05-1962	M	F	18-03-2011	
16	16	Marketing	Manager	19-03-1975	S	M	20-12-2007	
17	17	Marketing	Assistant	03-05-1987	S	M	26-01-2007	
18	18	Marketing	Specialist	06-03-1978	S	M	07-02-2011	
19	19	Marketing	Assistant	29-01-1978	S	F	14-02-2011	
20	20	Marketing	Assistant	19-09-1979	M	F	07-01-2011	
21	21	Marketing	Specialist	04-02-1980	M	M	02-03-2009	
22	22	Marketing	Specialist	21-05-1987	S	M	12-12-2008	
23	23	Marketing	Specialist	13-09-1982	M	F	12-01-2009	
24	24	Marketing	Specialist	18-06-1979	M	F	18-01-2009	
25	25	Vice	President of Production	07-01-1983	S	M	03-02-2009	
26	26	Production	Control Manager	09-11-1982	M	M	01-12-2008	
27	27	Production	Supervisor - WC60	08-10-1956	S	F	27-02-2008	
28	28	Production	Technician - WC60	13-03-1988	M	M	30-06-2006	
29	29	Production	Technician - WC60	23-09-1979	S	M	23-01-2009	
30	30	Production	Technician - WC60	28-09-1989	M	F	29-01-2009	
31	31	Production	Technician - WC60	20-05-1986	M	F	04-01-2009	
32	32	Production	Technician - WC60	20-07-1977	M	F	29-12-2008	
33	33	Production	Technician - WC60	26-12-1976	M	M	17-12-2008	
34	34	Production	Technician - WC60	20-06-1987	M	F	16-02-2009	
35	35	Production	Technician - WC60	10-01-1977	M	M	08-02-2009	
36	36	Production	Technician - WC60	11-07-1984	M	M	10-02-2009	
37	37	Production	Technician - WC60	07-08-1986	S	M	07-03-2009	
38	38	Production	Technician - WC60	11-12-1997	S	F	17-03-2009	

12 COLUMNS, 280 ROWS. Column profiling based on top 1000 rows.

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STEP 6: Create Age Column (Transform Step 3)

1. Go to: Add Column → Custom Column
2. Formula: Age = Date.Year(DateTime.LocalNow()) - Date.Year([BirthDate])

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Queries [2]

Department_Table

HumanResources.Employee

Table: RenameColumns(ChangedType1,({"JobTitle.2", "Role"}, {"JobTitle.1", "Department"}))

BusinessEntityID	OrganizationLevel	Department	Role	BirthDate	MaritalStatus	Gender	HireDate	SalaryFlag
1	0	Chief	Executive Officer	29-01-1969	S	M	14-01-2009	
2	1	Vice	President of Engineering	05-09-1972	S	F	31-01-2008	
3	2	Engineering	Manager	12-12-1974	M	M	12-12-2007	
4	3	Senior	Tool Designer	29-12-1974	S	M	05-12-2007	
5	3	Design	Engineer	27-09-1992	M	F	06-01-2008	
6	3	Design	Engineer	12-03-1959	M	M	24-01-2008	
7	3	Research and Development	Manager	24-02-1987	M	M	08-02-2009	
8	4	Research and Development	Engineer	05-06-1986	S	F	29-12-2008	
9	4	Research and Development	Manager	21-03-1979	M	F	18-01-2009	
10	10	Research and Development	Manager	30-12-1984	M	M	03-05-2009	
11	11	Senior	Tool Designer	17-01-1978	S	M	05-12-2010	
12	12	Tool	Designer	29-07-1959	M	M	11-12-2007	
13	13	Tool	Designer	28-09-1989	M	F	29-12-2010	
14	14	Senior	Design Engineer	18-06-1979	S	M	30-12-2010	
15	15	Design	Engineer	03-05-1962	M	F	18-03-2011	
16	16	Marketing	Manager	19-03-1975	S	M	20-12-2007	
17	17	Marketing	Assistant	03-05-1987	S	M	26-01-2007	
18	18	Marketing	Specialist	06-03-1978	S	M	07-02-2011	
19	19	Marketing	Assistant	29-01-1978	S	F	14-02-2011	
20	20	Marketing	Assistant	19-09-1979	M	F	07-01-2011	
21	21	Marketing	Specialist	04-02-1980	M	M	02-03-2009	
22	22	Marketing	Specialist	21-05-1987	S	M	12-12-2008	
23	23	Marketing	Specialist	13-09-1982	M	F	12-01-2009	
24	24	Marketing	Specialist	18-06-1979	M	F	18-01-2009	
25	25	Vice	President of Production	07-01-1983	S	M	03-02-2009	
26	26	Production	Control Manager	09-11-1982	M	M	01-12-2008	
27	27	Production	Supervisor - WC60	08-10-1956	S	F	27-02-2008	
28	28	Production	Technician - WC60	13-03-1988	M	M	30-06-2006	
29	29	Production	Technician - WC60	23-09-1979	S	M	23-01-2009	
30	30	Production	Technician - WC60	28-09-1989	M	F	29-01-2009	
31	31	Production	Technician - WC60	20-05-1986	M	F	04-01-2009	
32	32	Production	Technician - WC60	20-07-1977	M	F	29-12-2008	
33	33	Production	Technician - WC60	26-12-1976	M	M	17-12-2008	
34	34	Production	Technician - WC60	20-06-1987	M	F	16-02-2009	
35	35	Production	Technician - WC60	10-01-1977	M	M	08-02-2009	
36	36	Production	Technician - WC60	11-07-1984	M	M	10-02-2009	
37	37	Production	Technician - WC60	07-08-1986	S	M	07-03-2009	
38	38	Production	Technician - WC60	11-12-1997	S	F	17-03-2009	

12 COLUMNS, 280 ROWS. Column profiling based on top 1000 rows.

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Custom Column

Add a column that is computed from the other columns.

New column name: Custom

Custom column formula (fx):

Available columns:

- BusinessEntityID
- OrganizationLevel
- Department
- Role
- BirthDate
- MaritalStatus
- Gender

<< Insert

Learn about Power Query formulas

✓ No syntax errors have been detected.

OK Cancel

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Column From Custom Examples Column Function Duplicate Column

Conditional Column Index Column Formula Merge Columns Extract From Text From Number From Date & Time

Statistics Standard Scientific Rounding Information Date Time Duration

Queries [2]

Department_Table

HumanResources.Employee

Table.RenameColumns(*Changed Type1,({"JobTitle.2", "Role"}, {"JobTitle.1", "Department"}))

BusinessEntityID OrganizationLevel Department Role BirthDate MaritalStatus Gender HireDate SalaryFlag

1 1 1 0 Chief Executive Officer 29-01-1969 S M 14-01-2009

2 2 2 1 Vice President of Engineering 01-08-1971 S F 31-01-2008

3 3 3 2 Engineering Manager 12-11-1974 M M 11-11-2007

4 4 4 3 Senior Tool Designer 22-12-1974 S M 05-12-2007

5 5 5 3 Design 06-01-2008

6 6 6 3 Design 24-01-2008

7 7 7 3 Research 08-02-2009

8 8 8 4 Research 29-12-2008

9 9 9 4 Research 18-01-2009

10 10 10 4 Research 01-05-2009

11 11 11 3 Senior 05-12-2010

12 12 12 4 Tool 11-12-2007

13 13 13 3 Senior 29-12-2010

14 14 14 3 Senior 30-12-2010

15 15 15 3 Design 18-03-2011

16 16 16 1 Marketing 20-12-2007

17 17 17 2 Marketing 26-01-2007

18 18 18 2 Marketing 07-02-2011

19 19 19 2 Marketing 14-02-2011

20 20 20 2 Marketing 07-03-2011

21 21 21 2 Marketing 02-03-2009

22 22 22 2 Marketing 12-12-2008

23 23 23 2 Marketing 12-01-2009

24 24 24 2 Marketing 18-01-2009

25 25 25 1 Vice 01-02-2009

26 26 26 2 Production 01-12-2008

27 27 27 3 Production Supervisor - WC60 08-10-1956 S F 27-02-2008

28 28 28 4 Production Technician - WC60 13-03-1988 M M 30-06-2006

29 29 29 4 Production Technician - WC60 25-09-1979 S M 23-01-2009

30 30 30 4 Production Technician - WC60 28-09-1989 M F 29-01-2009

31 31 31 4 Production Technician - WC60 20-05-1986 M F 04-01-2009

32 32 32 4 Production Technician - WC60 20-07-1977 M F 29-12-2008

33 33 33 4 Production Technician - WC60 20-12-1976 M M 17-12-2008

34 34 34 4 Production Technician - WC60 10-06-1987 M F 16-02-2009

35 35 35 4 Production Technician - WC60 10-01-1977 M M 08-02-2009

36 36 36 4 Production Technician - WC60 31-07-1984 M M 10-02-2009

37 37 37 4 Production Technician - WC60 07-08-1980 S M 07-03-2009

12 COLUMNS, 290 ROWS Column profiling based on top 1000 rows

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File Home Transform Add Column View Tools Help

Column From Custom Examples Column Function Duplicate Column

Conditional Column Index Column Formula Merge Columns Extract From Text From Number From Date & Time

Statistics Standard Scientific Rounding Information Date Time Duration

Queries [2]

Department_Table

HumanResources.Employee

Table.AddColumn(*Renamed Columns*, "Age", each Date.Year(DateTime.LocalNow()) - Date.Year([BirthDate]))

BirthDate MaritalStatus Gender HireDate SalaryFlag Age VacationHours SickLeaveHours CurrentFlag

1 29-01-1969 S M 14-01-2009 39 0 0 1

2 01-08-1971 S F 31-01-2008 37 1 20 1

3 12-11-1974 M M 11-11-2007 35 21 3 1

4 22-12-1974 S M 05-12-2007 34 80 3 1

5 27-09-1982 M F 06-01-2008 32 5 22 1

6 11-03-1959 M M 24-01-2008 50 6 23 1

7 24-02-1987 M M 08-02-2009 32 83 50 1

8 05-06-1965 S F 29-12-2008 54 62 51 1

9 22-01-1979 M F 16-01-2009 38 83 51 1

10 30-11-1984 M M 03-05-2009 35 16 64 1

11 17-01-1978 S M 05-12-2010 31 7 23 1

12 29-07-1955 M M 12-12-2007 53 9 24 1

13 28-05-1989 M F 23-12-2010 30 8 24 1

14 16-06-1979 S M 30-12-2010 31 3 21 1

15 01-05-1961 M F 18-01-2011 51 4 13 1

16 19-03-1975 S M 20-12-2007 35 40 40 1

17 01-05-1987 S M 24-01-2007 36 42 41 1

18 06-03-1978 S F 07-02-2011 36 48 44 1

19 29-01-1978 S F 14-02-2011 37 43 41 1

20 17-03-1975 M F 07-03-2011 38 41 40 1

21 04-02-1986 M M 02-02-2009 34 44 42 1

22 11-01-1987 S M 12-12-2008 37 43 41 1

23 11-08-1962 M F 12-01-2009 56 46 43 1

24 18-08-1979 M F 18-01-2009 41 47 43 1

25 07-01-1983 S M 03-02-2009 36 64 52 1

26 01-11-1982 M M 01-12-2008 37 43 41 1

27 08-10-1956 S F 27-02-2008 60 80 80 1

28 13-03-1988 M M 30-06-2006 31 21 30 1

29 25-09-1979 S M 23-01-2009 39 19 29 1

30 28-09-1989 M F 29-01-2009 34 14 27 1

31 20-05-1986 M F 04-01-2009 38 18 29 1

32 10-07-1977 M F 29-12-2008 37 23 31 1

33 20-12-1976 M M 17-12-2008 38 27 28 1

34 10-06-1987 M F 16-02-2009 36 15 27 1

35 10-01-1977 M M 08-02-2009 37 22 31 1

36 31-07-1984 M M 10-02-2009 35 20 30 1

37 07-08-1980 S M 07-03-2009 38 16 28 1

13 COLUMNS, 290 ROWS Column profiling based on top 1000 rows

REVIEW DOWNLOADED AT 1843

STEP 7: Fix Data Types (Transform Step 4)

1. Set:

BusinessEntityID → Whole Number

BirthDate → Date

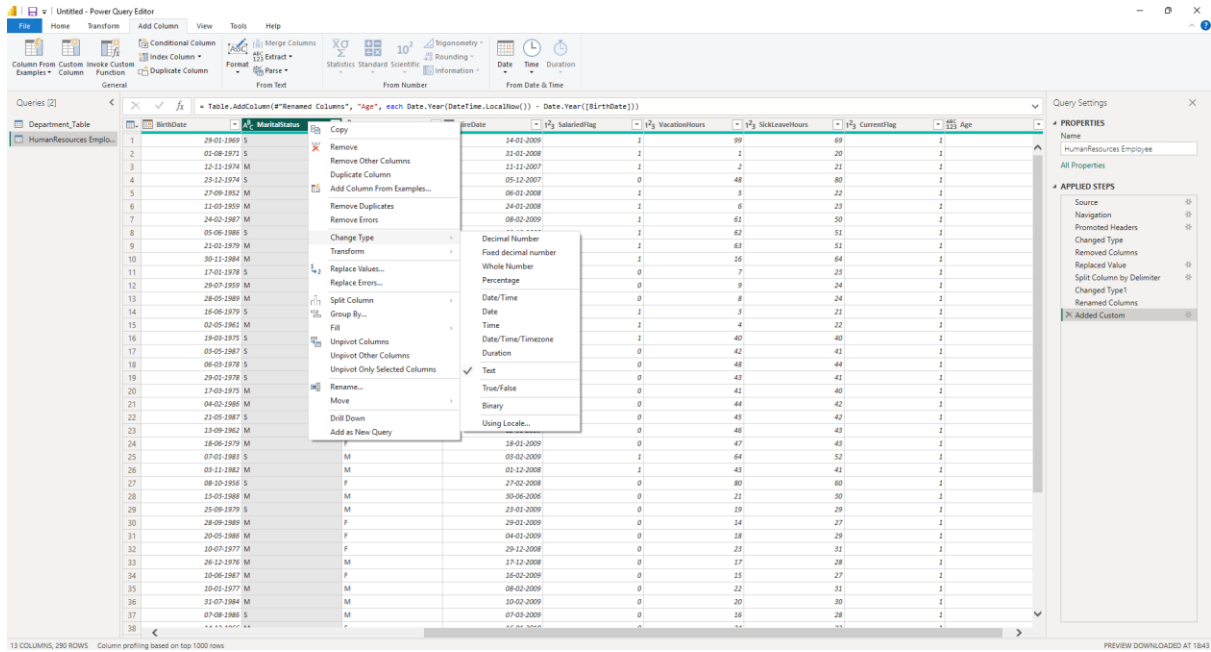
HireDate → Date

VacationHours → Whole Number

SickLeaveHours → Whole Number

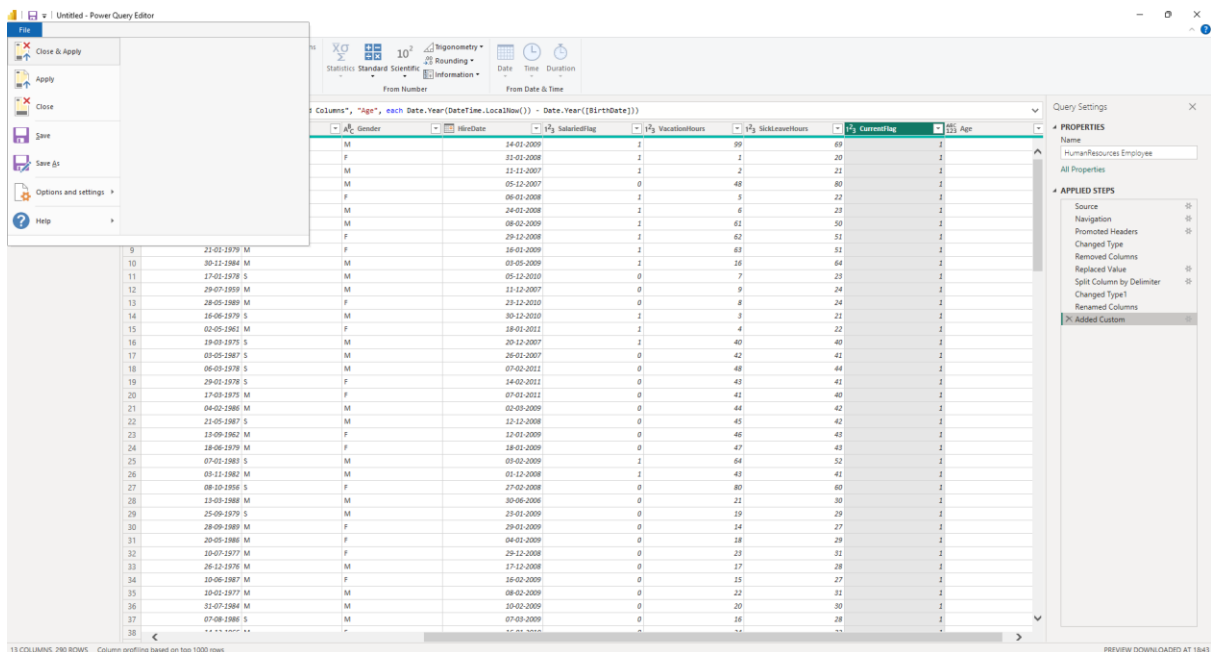
Gender → Text

JobTitle → Text



STEP 8: Load Data (L in ETL)

1. Click: **Close & Apply**
2. Data is now loaded into Power BI model

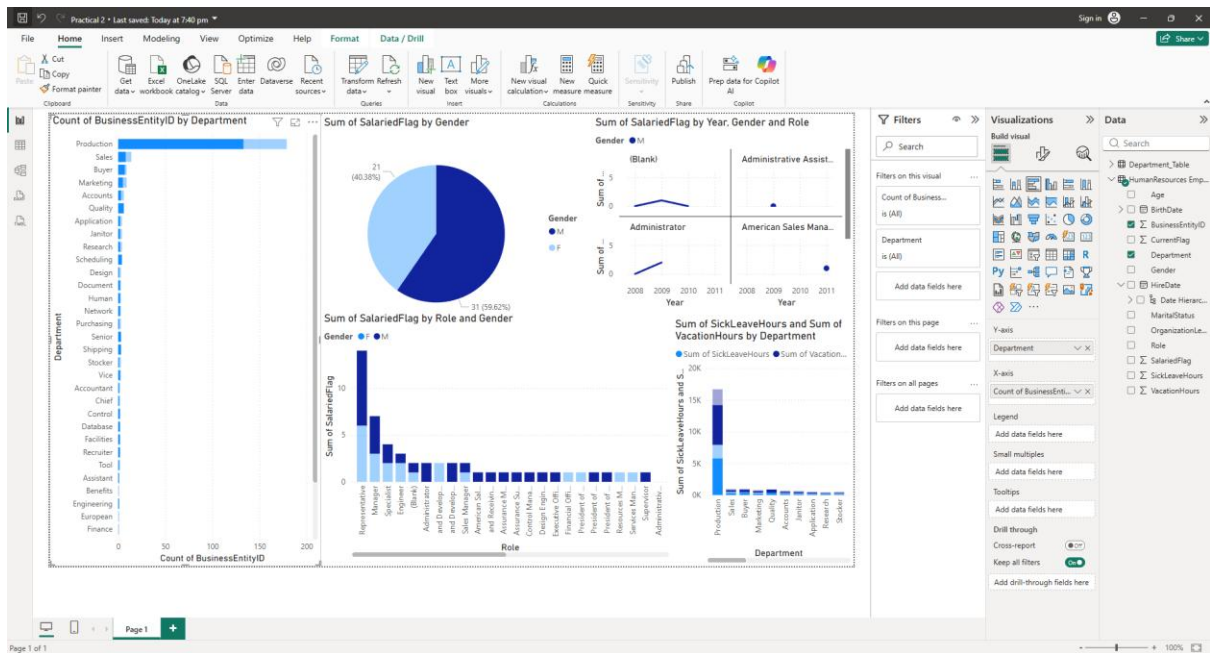


STEP 9: Create Simple Output (Proof)

1. Add a **Table Visual**
2. Show:
 - BusinessEntityID

- JobTitle
- Gender
- HireDate
- VacationHours
- SickLeaveHours

This is final ETL output



STEP 10: Save File

1. File → Save
2. Save as .pbix

